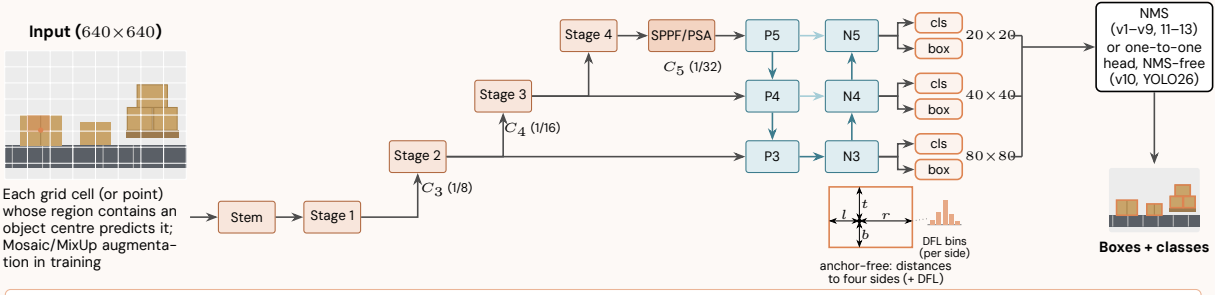
Backbone: Darknet-19/53 (v2–v3) → CSPDarknet (v4, v5) → E-ELAN (v7) → C2f (v8) → GELAN (v9) → C3k2 + C2PSA (11) → R-ELAN + area attention (v12) → HyperACE (v13)

Neck: FPN top-down + PAN bottom-up; Rep-PAN (v6), FullPAD (v13)

Head: coupled, anchor-based (v2–v5, v7) → decoupled, anchor-free (YOLOX, v6, v8+)



Training. *Label assignment*: centre cell / max IoU (v1–v3) → multi-anchor (v4, v5) → SimOTA (YOLOX) → TAL, Eq. (8) (v6, v8–v13) → consistent dual one-to-many + one-to-one (v10) → STAL (YOLO26). *Box loss*: SSE → CIoU, Eq. (10) → CIoU + DFL, Eq. (11) → DFL-free with ProgLoss (YOLO26). *Classification*: BCE or VFL. *Optimiser*: SGD → MuSGD (YOLO26). *Deployment*: re-parameterisation (v6, v7), INT8 export; see Table 4.